

PERUGIA, Italy

WMO: 161810

Lat: 43.0925N

Lon: 12.5047E

Elev: 669

StdP: 14.34

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.6	28.2	17.6	13.7	34.3	20.8	16.0	33.5	25.6	41.9	22.6	43.1	4.9	0	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.2	95.4	67.9	93.2	67.6	91.0	67.5	71.6	86.5	70.4	85.5	69.3	84.5	8.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.4	103.2	75.8	66.1	98.4	75.4	64.5	93.1	74.2	35.7	86.9	34.7	85.9	33.7	84.9	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.5	18.6	16.4	20.5	100.8	2.3	2.0	18.8	102.2	17.5	103.4	16.3	104.6	14.6	106.1
			19.9	73.8	2.3	1.8	18.2	75.1	16.9	76.2	15.6	77.2	13.9	78.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.8	41.6	43.4	48.5	54.3	61.3	70.3	75.8	68.0	59.9	50.1	43.2
		DBStd	13.31	5.20	5.45	5.47	5.14	5.04	5.69	4.33	4.43	5.37	5.43	6.68
		HDD50	883	265	191	93	19	0	0	0	0	5	80	229
		HDD65	3640	727	604	511	321	136	17	0	1	31	169	446
		CDD50	3720	3	7	48	149	352	610	799	541	311	84	17
		CDD65	1002	0	0	0	1	23	177	335	334	122	10	0
		CDH74	11763	0	0	0	37	364	2170	4156	3812	1142	82	0
		CDH80	5354	0	0	0	3	72	929	2095	1907	339	9	0
(14)	Wind	WSAvg	6.5	6.1	7.2	7.6	6.7	6.3	6.8	6.2	6.8	6.3	5.9	6.0
(15)	Precipitation	PrecAvg	32.0	2.0	2.3	2.4	2.9	2.4	2.2	1.5	1.9	3.5	3.8	4.1
(16)		PrecMax	45.3	5.9	5.4	5.3	4.7	5.9	4.8	4.4	5.3	7.7	8.7	8.3
(17)		PrecMin	18.8	0.4	0.2	0.2	0.7	1.1	0.6	0.0	0.0	1.5	0.3	0.1
(18)		PrecStd	5.9	1.3	1.4	1.3	1.0	1.2	1.0	0.9	1.2	1.3	1.9	2.3
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	57.8	62.8	71.3	78.6	86.4	96.4	98.9	98.9	91.6	80.6	69.4	61.2
(20)		0.4% MCWB	51.5	51.1	53.6	60.7	64.9	68.5	67.4	67.7	66.5	63.4	60.8	56.4
(21)		2% DB	55.3	58.9	67.7	73.5	81.9	92.9	96.5	96.4	87.8	76.6	65.4	58.7
(22)		2% MCWB	49.8	49.7	53.2	57.9	63.5	67.2	67.1	68.2	65.7	62.5	58.9	54.8
(23)		5% DB	53.2	55.6	63.0	69.9	78.5	89.6	93.3	93.1	84.1	73.3	62.5	55.8
(24)		5% MCWB	48.4	48.7	52.0	56.8	62.5	67.7	67.5	68.1	65.1	61.9	57.2	52.3
(25)		10% DB	50.3	53.4	59.7	66.4	74.9	86.1	91.0	89.8	80.6	70.1	59.4	53.6
(26)		10% MCWB	46.9	47.4	50.7	54.8	61.0	66.6	67.1	67.9	63.8	61.1	55.0	50.3
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	53.1	53.9	56.8	62.6	67.4	72.0	72.8	73.8	70.5	66.4	63.1	57.4
(28)		0.4% MCDB	56.8	59.2	65.6	74.0	81.9	87.6	87.5	87.3	82.0	72.7	67.2	60.4
(29)		2% WB	50.8	51.7	54.9	59.7	65.2	70.1	71.1	71.9	68.4	64.4	59.9	54.9
(30)		2% MCDB	53.9	56.1	63.2	69.9	78.1	86.9	86.4	86.7	80.6	71.6	64.0	57.9
(31)		5% WB	49.4	49.9	53.1	57.7	63.6	68.7	70.1	70.6	67.2	63.1	57.6	53.1
(32)		5% MCDB	52.0	53.9	60.9	67.3	75.5	85.4	85.7	85.3	79.1	70.5	61.4	56.0
(33)		10% WB	47.5	48.2	51.7	56.0	61.9	67.3	68.9	69.5	65.8	61.9	55.7	50.8
(34)		10% MCDB	50.0	52.2	58.7	64.5	72.2	82.8	84.7	84.2	76.9	68.9	59.6	53.2
(35)	Mean Daily Temperature Range	MDBR	14.0	17.2	19.7	21.1	22.1	25.2	27.2	26.1	22.2	18.6	15.0	12.9
(36)		5% DB MCDBR	17.0	21.6	27.1	28.5	28.6	31.0	31.6	30.8	27.8	22.6	17.1	13.6
(37)		5% WB MCWBR	12.2	13.9	15.2	15.1	13.4	10.5	9.6	9.6	9.9	11.4	11.0	10.3
(38)		5% DB MCDBR	13.8	17.8	22.6	23.9	25.2	28.3	27.0	26.0	22.3	18.5	15.1	12.0
(39)		5% WB MCWBR	10.6	12.3	13.3	13.4	12.7	10.3	9.3	9.5	9.1	10.3	10.3	9.6
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.356	0.391	0.428	0.432	0.430	0.418	0.416	0.404	0.387	0.361	0.333
(41)		taud	2.429	2.342	2.253	2.165	2.197	2.249	2.284	2.307	2.326	2.359	2.371	2.443
(42)		Ebn at Noon	252	264	267	266	268	268	270	267	260	249	237	240
(43)		Edh at Noon	24	31	38	45	45	43	41	39	36	30	26	22
(44)	All-Sky Solar Radiation	RadAvg	489	749	1114	1481	1833	2121	2211	1918	1374	907	545	436
(45)		RadStd	60	94	154	143	176	109	116	130	83	89	58	56

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	+148	N/A

Nomenclature: See separate page